A targeted operational reset

Where incident handling slows down, what the evidence shows and which changes are worth testing.

INCIDENTS	MEDIAN RESOLUTION	90TH PERCENTILE
24,918	22.1h	15.9d
Distinct incidents	23,362 valid durations	Extended tail of cases

Executive assessment

The incident records highlight three priorities: repeated handling, long-running cases and gaps in lifecycle data. These are sensible areas to investigate. The records alone cannot show which handoffs were avoidable or how much time a change would save.

01 Reduce rework

2,737 incidents had both at least three reassignments and seven or more handling events. Check whether ownership and escalation decisions led to avoidable work.

02 Tackle long cases

5,360 incidents exceeded seven days among cases with valid resolution times. Investigate where these cases spent time waiting.

03 Repair lifecycle data

1,556 incidents have no valid resolution duration. Check timestamps and strengthen record validation.

Recommended management decision

Approve a 12-week review and pilot, led jointly by IT operations, the service desk and platform owners. Begin by reviewing repeated-routing cases, checking who should own them and where delays occur. Then test agreed changes against a comparable group, with safeguards in place.

WHAT SUCCESS WOULD MEAN

Fewer avoidable handoffs and less repeated work, without compromising service quality. Track median and P90 resolution times, reassignments, routing returns, reopen rates and data completeness. These changes have not yet been implemented or tested.

Mandate, scope and evidence

The decision to make, the information available and the limits of the analysis.

Decision question

Which incident-management bottlenecks should the IT department prioritise for process improvement and targeted automation to reduce resolution delays, minimise unnecessary handling, and improve operational efficiency?

Evidence base

INCIDENT RECORDS	EVENT RECORDS	VALID DURATIONS
24,918	**141,712**	**23,362**
One record per incident	Historical event log	93.76% of incidents

The analysis checked incident IDs and event order, prepared incident-level summaries and calculated key measures in SQL. Python was used to test patterns in the data. Three Power BI dashboards present the findings, supported by 28 outputs recorded in the Colab evidence register.

KPIs monitored

Outcome metric	Management use
Incident volume / throughput	Track incident demand, recorded throughput and data coverage.
Median and P90 resolution	Monitor typical service time and the long-delay tail.
Reassignment / handling events	Spot cases that may involve repeated work or handoff delays.
Reopen rate	Guard against faster closure at the expense of quality.

Limits of the evidence: The dataset is historical and anonymised; service-level targets, staffing rosters, transfer reasons, cost rates and results from live automation were not provided. SLA compliance cannot be assessed without the agreed thresholds.

The performance picture

Long-running cases and gaps in resolution data matter more than volume alone.

Core performance indicators

REASSIGNED

45.63%

11,369 incidents

REOPENED

1.10%

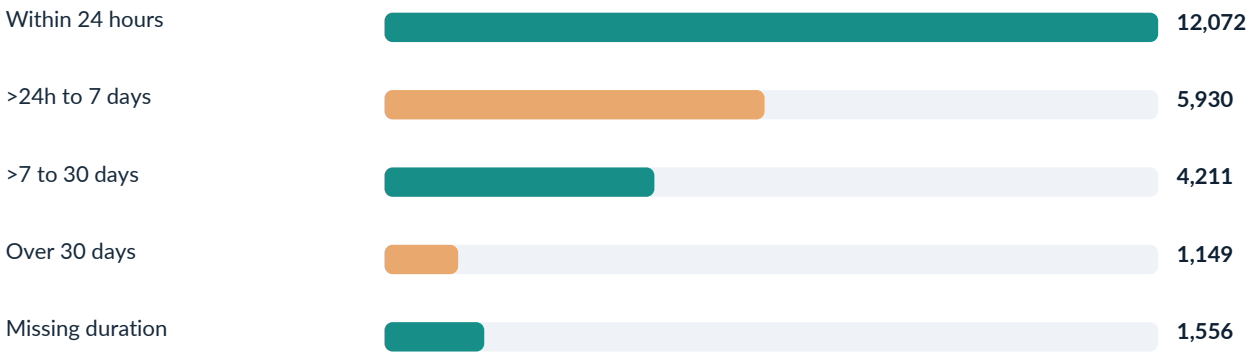
275 incidents

EVENTS / INCIDENT

5.69

141,712 events

Resolution distribution (24,918 incidents)



What the distribution means

22.94% Over seven days

5,360 of 23,362 valid records exceeded a week. Review waiting time and escalation steps in these cases.

4.92% Over thirty days

1,149 of 23,362 valid records exceeded a month, making long delays a material issue.

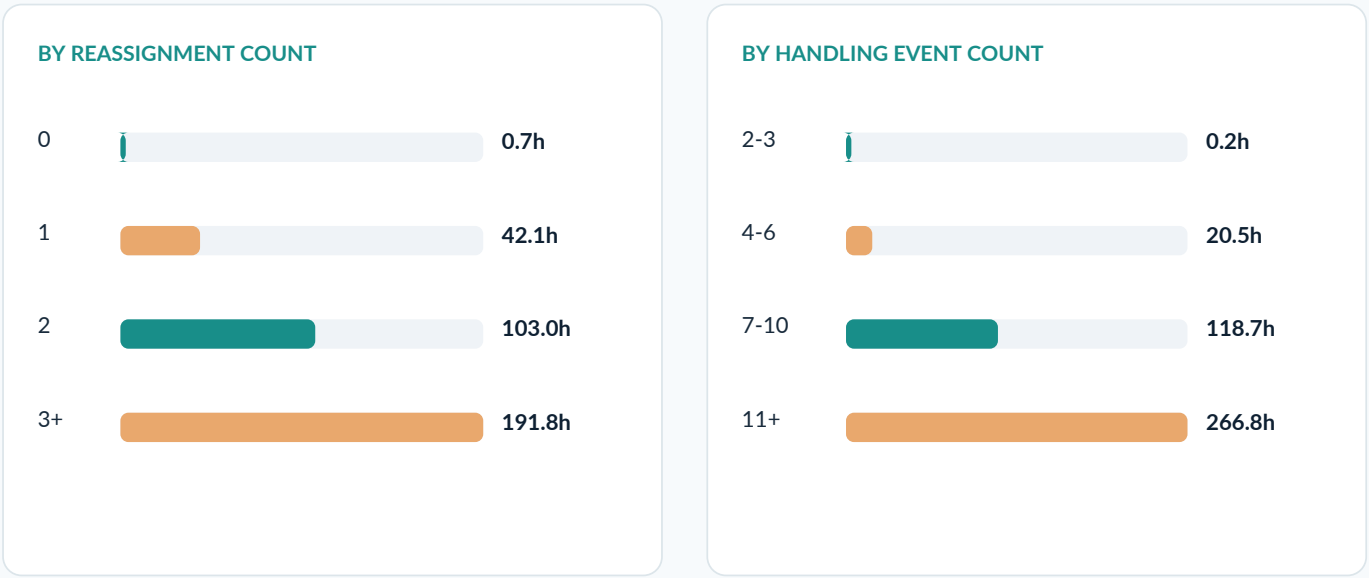
DATA COMPLETENESS MATTERS

For 1,556 incidents (6.24%), a valid resolution duration is unavailable. All 24,918 incidents have a final recorded state of Closed, so missing durations must not be treated as proof that tickets are still open. Check the source timestamps and lifecycle records.

Handling intensity tracks delay

Incidents with more handling activity tend to take longer; this does not establish the cause.

Median resolution time by cohort



What the Python tests add

Diagnostic	Validated result	Interpretation
Kruskal-Wallis - reassignments	H = 6,493.55; epsilon-squared = 0.2779	Resolution times differ markedly between the groups.
Kruskal-Wallis - handling	Epsilon-squared = 0.4820	Event-count groups show an even clearer difference.
High-overlap cohort	2,737 incidents; median 11,657 min	At least three reassignments and seven handling events.

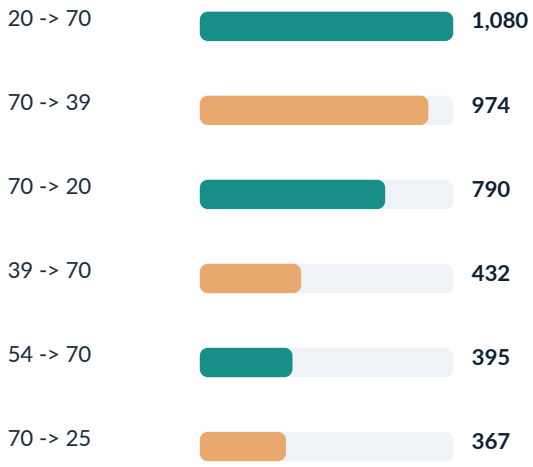
Management implication: Flag cases with seven or more handling events or three or more reassignments for review. Cases meeting both thresholds deserve particular attention. These are screening rules, not proven causes of delay or SLA breaches.

Transfers and returns

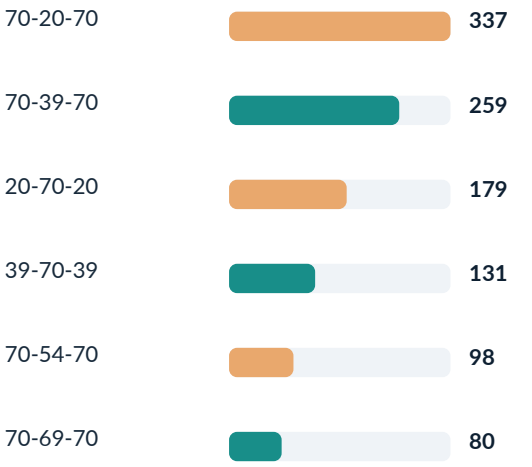
A few support-group relationships account for repeated transfers and are worth reviewing first.

GROUP CHANGES	ROUTING RETURNS	TOP-FOUR RETURNS
17,914	3,231	906
9,224 incidents	1,963 incidents	28.0% of return events

MOST FREQUENT TRANSFERS



MOST FREQUENT RETURNS



How the routing sequence was reconstructed

Python compared 101,409 adjacent event pairs with known assignment groups and found 17,914 changes (17.67%). It also identified 3,231 A-to-B-to-A returns across 1,963 incidents. One incident can contain several returns. The four most common return patterns involve Groups 70, 20 and 39.

Audit ownership loops

Review a sample involving Groups 70, 20 and 39. Separate necessary escalations from avoidable returns.

Preserve sequencing caveats

14,213 events lack known assignment groups; 4,216 observed changes occur with tied timestamps.

Association is not causation

Routing returns are linked to longer cases in the raw data, but the picture changes after adjustment.

Unadjusted routing-return comparison

WITH A RETURN

10,805m

n = 1,863 valid cases

NO RETURN

882m

n = 20,362 valid cases

RANK-BISERIAL

-0.5714

Mann-Whitney; p < 0.001

The difference is large, but cases with returns may also be more complex and involve more activity. The comparison includes only records with the necessary data. It does not measure the time that could be saved by preventing a return.

Compare more similar handling cohorts

Event band	No return median	Return median	Statistical diagnostic
7-10 events	7,133m (n=3,965)	7,326m (n=769)	U = 1,439,252.5; p = 0.0139; rb = -0.0559
11+ events	14,658m (n=1,227)	18,463m (n=941)	U = 506,291; p < 0.001; rb = -0.1230

Regression sensitivity check

An OLS model with robust standard errors used 22,225 complete records to examine log-transformed resolution time. Its R-squared was 0.4882. The adjusted coefficient for log event count was positive (+5.6823), while those for reassignment count (-0.1790) and a routing return (-0.2754) were negative.

Interpretation: The adjusted results differ from the raw comparison. Case complexity, related predictors and modelling choices may affect the findings. We cannot conclude that routing returns cause delays or that removing them will save time.

What the evidence can tell us

Use the four working hypotheses to guide what needs checking with the service teams.

Hypotheses and conclusions

Working hypothesis	Assessment	Basis and management interpretation
H1 Reassignment bottlenecks	Supported descriptively	Resolution-time distributions vary markedly with reassignment count; validate the reasons for each transfer.
H2 Repeated handling	Supported descriptively	Higher event-count groups have much longer medians; test whether extra events represent rework or legitimate complexity.
H3 Demand vs throughput	Inconclusive	Recorded monthly throughput peaked at 7,794, then fell. The available data cannot explain whether demand, coverage or timing drove this change.
H4 Priority x complexity	Partially explored	The analysis does not isolate the effect of incident priority. A clearer breakdown needs additional operational context.

Root-cause validation agenda

Operational interview evidence

Inspect transfer reasons, ownership rules, wait states, service categories and staffing coverage alongside actual incident histories.

Technical and data evidence

Check event order, timestamp rules, SLA definitions, workflow settings and access controls.

Practical, prioritised recommendations

Five practical actions, with clear owners, evidence to collect and safeguards.

Intervention portfolio

Priority / owner	Recommended action	Proof to collect	Control / safeguard
P1 Service desk lead	Review 30 to 50 repeat-routing cases involving Groups 70, 20 and 39. Record why each transfer happened and whether it was needed.	Categorised causes, ownership map, avoidable-loop rate.	Do not suppress required specialist escalation.
P1 Incident manager	Flag cases at seven handling events or three reassignments for a supervisor review of ownership, blockers and next steps.	Review completion rate; median/P90 by cohort.	Use a review alert, not automatic closure.
P1 Platform/data owner	Investigate 1,556 records without valid durations; check source timestamps, add validation and monitor exceptions.	Exception log, reasons for gaps, duration completeness.	Respect audit history; avoid fabricated timestamps.
P2 Process owner	Agree assignment criteria, a handoff checklist and clear escalation responsibilities for frequent transfer pairs.	Updated decision tree, staff sign-off, routing checks.	Keep a human override for atypical cases.
P2 Automation lead	Test a routing suggestion or checklist for well-understood cases, with staff retaining control of reassignments.	Accuracy tests, override rate, user feedback.	Least privilege, audit logs, rollback capability.

Sequencing principle: Confirm how the work should happen before automating it. Keep a named person responsible for exceptions.

A 12-week test-and-learn roadmap

A practical sequence for testing changes, measuring results and reversing them if needed.

Phased implementation

Weeks 1-2	**Baseline and sample** Choose eligible queues, confirm review thresholds, examine 30 to 50 cases, record why transfers happened and agree the baseline.
Weeks 3-4	**Co-design and controls** Agree an ownership map, handoff checklists and review alerts. Obtain team sign-off, confirm security controls and set a rollback plan.
Weeks 5-8	**Limited pilot** Test changes in selected queues with named owners. Track transfer reasons, handling activity, resolution times and staff overrides.
Weeks 9-12	**Evaluate and decide** Compare results with similar queues or a suitable control group. Review service quality and decide whether to adapt, expand or stop.

Proposed decision gates - not validated targets

Primary: fewer confirmed avoidable returns per eligible incident, compared with an agreed control or baseline. Supporting: fewer handling events in targeted cohorts and shorter median/P90 resolution. Guardrails: no increase in reopen rate, no deterioration in priority-specific service levels and no rise in unresolved exceptions or policy breaches.

A 10% relative reduction in confirmed avoidable returns can be used as an illustrative discussion threshold only after the client agrees definitions, baseline stability and sampling feasibility.

Seven questions for management

Proposed questions for a simulated client discussion; no interviews or answers are claimed.

01 Which stages of the incident-management process does management consider the most inefficient, and what evidence supports this assessment?

02 How are incidents currently assigned, transferred, and escalated between support teams, and what circumstances typically trigger reassignment?

03 How does management assess whether incident volumes, staffing capacity, and support-team workloads are appropriately balanced?

04 What are the organisation's expected service levels, resolution targets, and operational KPIs, and where does management believe performance falls short?

05 Which incident-management activities would management prioritise for process redesign or automation, and what operational, technical, or financial constraints must be considered?

06 Which repetitive incident-management tasks follow standard procedures, and what existing technologies or automation capabilities could support their improvement?

07 What challenges relating to system integration, data quality, access controls, and organisational readiness could affect the implementation of proposed process improvements or automation solutions?

Interview objective: replace inferred causes with operational facts, confirm the decision constraints and secure informed sponsorship for a time-limited pilot.

What must be resolved before change

Three important questions cannot be answered from historical records alone.

Missing context and how to close it

01 Incident-management workflow and root-cause context

Unknown: why teams reassigned incidents, whether returns were necessary, where incidents waited and whether capacity or specialist skills affected routing.
Close the gap: workflow walkthroughs, sampled ticket notes, team-lead interviews and a tagged transfer-reason taxonomy.

02 Performance targets and implementation constraints

Unknown: contractual service-level thresholds, service priorities, staffing capacity, budget, available licences and change-approval responsibilities.
Close the gap: confirm SLA definitions, queue ownership, operating constraints, feasible comparator groups and the pilot sponsor.

03 Data completeness, reliability and lifecycle visibility

Unknown: causes of 1,556 missing resolution durations, 14,213 events with unknown groups, tied timestamps and the effect of incomplete records.
Close the gap: audit the source system and event order, assign ownership for missing fields, and monitor exceptions.

Risks to the management decision

Risk	Mitigation
Unfounded causal claims	Use controlled comparison and case evidence, not observational correlations alone.
Unsafe automation	Keep human approval, least privilege, complete audit trails and a rollback path.

A measurable path to improvement

What management should approve, who should oversee it and how to measure the outcome.

Decision requested

Approve a focused pilot to improve incident routing and handling with an accountable sponsor, agreed measures, a small case-review sample and a comparable group. Set service-quality checks in advance. Delay broader automation until root causes and exception handling are understood.

Scorecard for operational governance

Metric	Baseline evidence	Proposed pilot use
Median / P90 resolution	22.1 hours / 15.9 days	Track all eligible incidents and targeted cohorts.
Reassignment rate	45.63% of incidents	Monitor trends; distinguish necessary and avoidable handoffs.
Observed routing returns	3,231 occurrences	Track validated avoidable-return rate and counts.
Reopen rate	1.10% of incidents	Service-quality safety guardrail.
Resolution-data completeness	93.76% valid durations	Aim to record valid durations for every new pilot case.
Handling intensity	5.69 events per incident	Check events per case and 7+ event cohort size.

Accountability and evidence trail

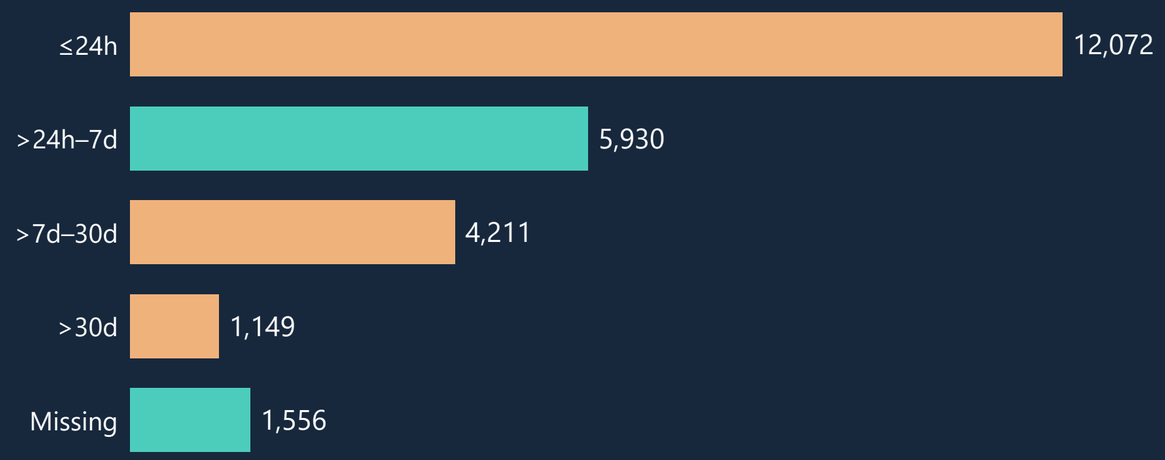
Executive sponsor: approve objectives, thresholds and pilot scope. Incident manager: own workflow reviews and team adoption. Platform owner: own data-quality and control design. Analytics lead: maintain the evaluation cohort and report change with confidence intervals where feasible. Risk/security: approve access, audit and rollback.

Evidence sources: the three Power BI dashboards reproduced after this report, validated incident and event summaries, SQL checks, Python analysis in Colab and the 28-entry evidence register. Findings are historical; actions and targets have not been implemented.

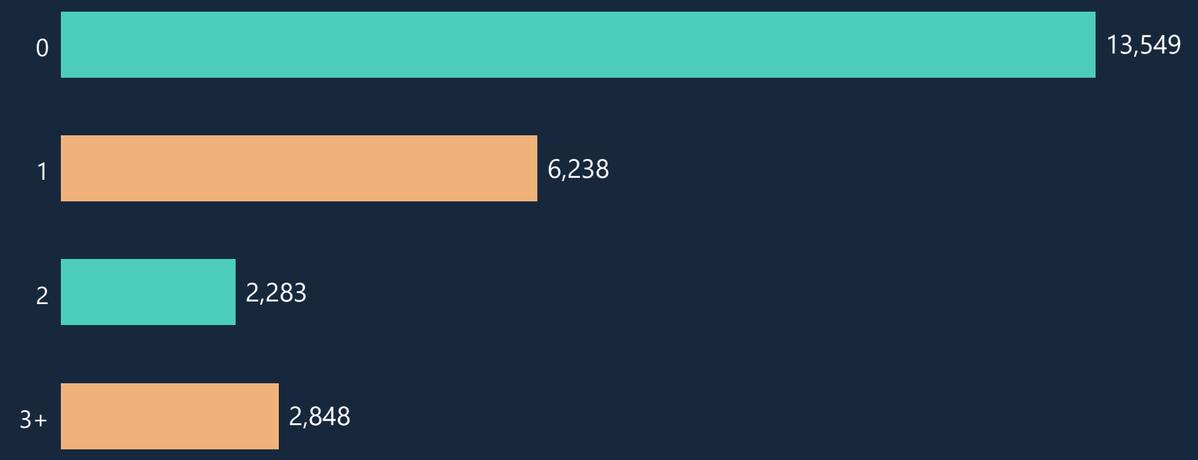
Incident Operations | Executive Overview

Total Incidents **24,918**	Median Resolution **22.1h**	P90 Resolution (Days) **15.9d**
Reassignment Rate **45.63%**	Reopen Rate **1.10%**	Average Events per Incident **5.69**

Incident Resolution Distribution



Incident Reassignment Distribution



01 REDUCE REPEATED HANDLING

2,737 incidents were reassigned at least 3 times and had 7 or more handling events.

Recommended action: Review these cases to identify unnecessary transfers, repeated work and unclear team responsibilities.

02 INVESTIGATE SLOW RESOLUTION

5,360 incidents took more than 7 days to resolve — 22.94% of cases with valid resolution times.

Recommended action: Identify where delays occur and prioritise improvements to the slowest incident-handling processes.

03 IMPROVE DATA RELIABILITY

1,556 incidents have no valid recorded resolution time.

Recommended action: Check whether these incidents remain unresolved or have missing timestamps, and improve record validation.

Historical data | 24,918 incidents | Descriptive analysis | Recommendations proposed, not implemented

Incident Operations | Resolution & Process Efficiency

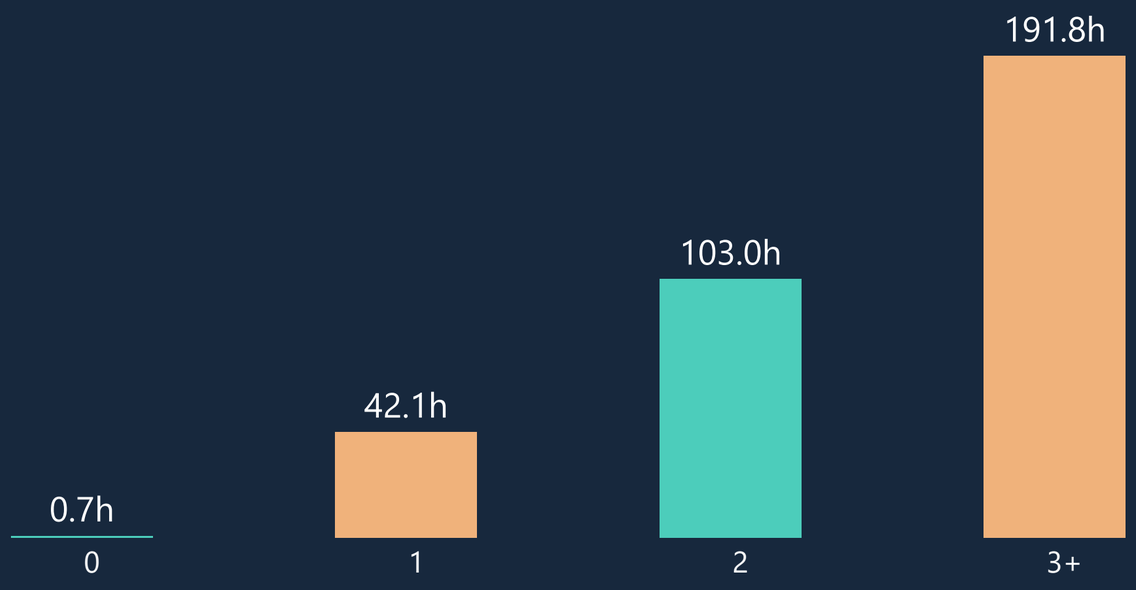
Valid Resolutions
23,362
93.76% of incidents

Over 7 Days
5,360
22.94% of valid

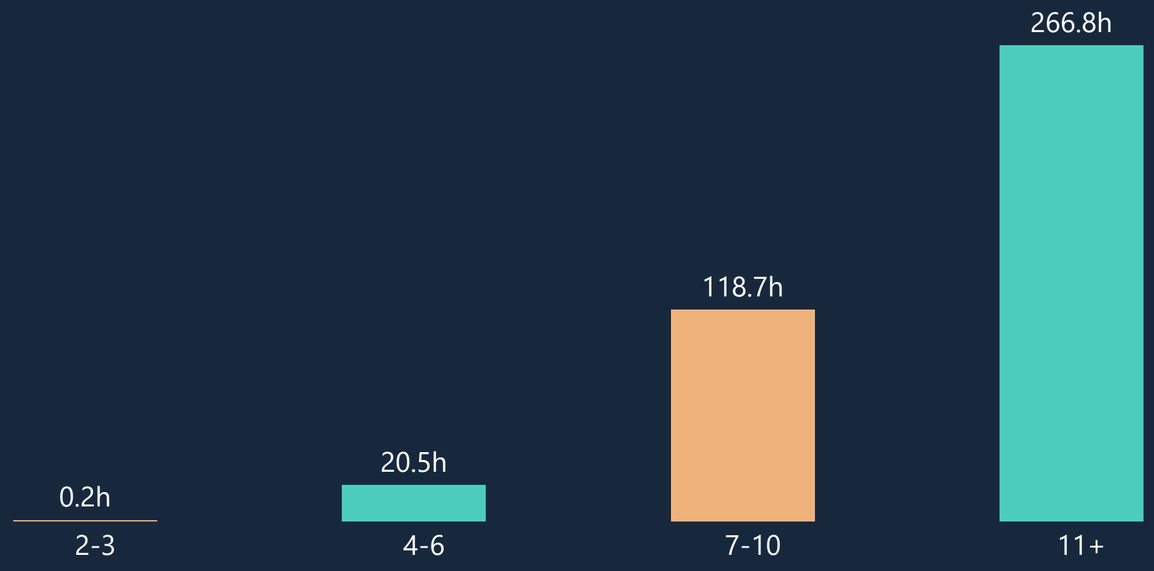
Over 30 Days
1,149
4.92% of valid

Missing Durations
1,556
6.24% of incidents

Median Resolution by Reassignment Group



Median Resolution by Handling Events



KEY FINDING — PROLONGED HANDLING

More handling activity is associated with longer resolution times.

Median resolution rises from **0.2 hours** (2–3 events) to **266.8 hours** (11+ events).

PROPOSED ACTION — EARLY ESCALATION REVIEW

Pilot a review when an incident reaches **7 handling events or 3 reassignments**.

Check case ownership, waiting times and escalation delays. Monitor median/P90 resolution and reopen rates.

23,362 valid durations | 1,556 missing | Observed associations, not causation | Actions proposed

Incident Operations | Routing & Transformation Opportunities

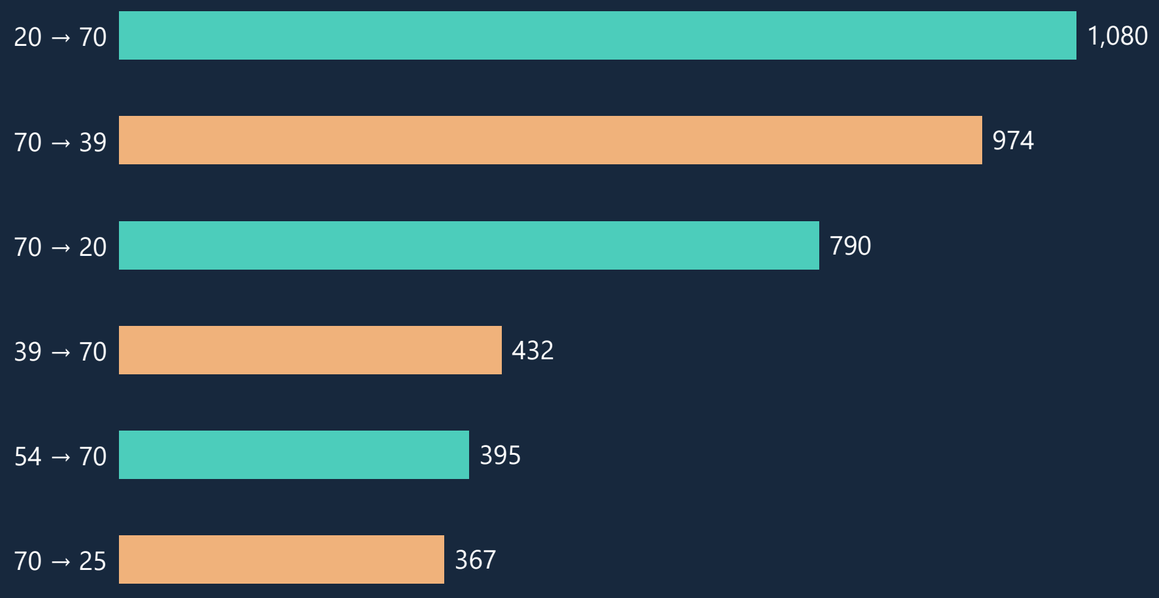
Support-Group Changes
17,914
Observed transfers

Incidents transferred
9,224
37.02% of incidents

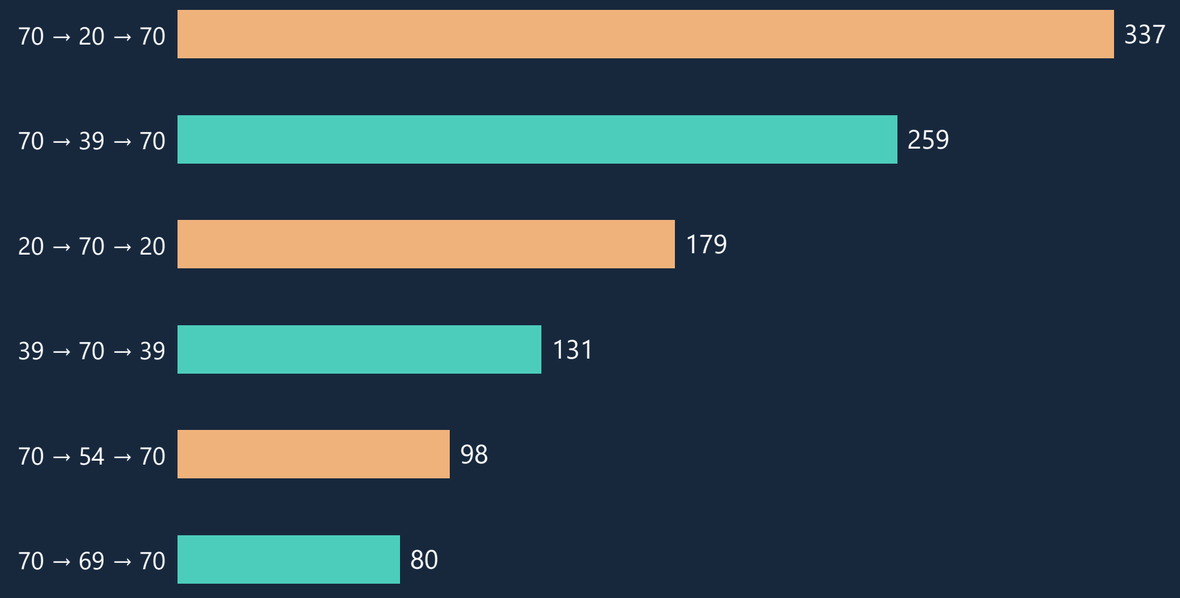
Routing Returns
3,231
A → B → A patterns

Incidents with Returns
1,963
7.88% of incidents

Most Frequent Support-Team Transfers



Most Frequent Routing Returns



KEY FINDING — REPEATED ROUTING

906 routing returns (28.0%) occurred across the four leading patterns involving Groups 70, 20 and 39.

These recurring transfers warrant investigation.

PROPOSED ACTION — IMPROVE ROUTING

Review 30–50 incidents from these recurring patterns with the relevant team leads.

Identify unnecessary handoffs, clarify ownership and pilot improved assignment rules where justified.

Historical data | 14,213 events with unknown assignment groups | 4,216 changes with tied timestamps | Proposed actions only